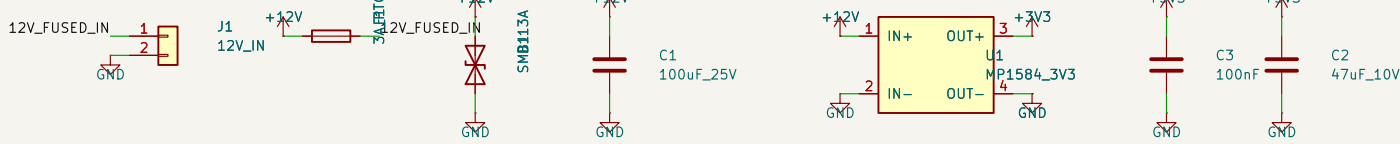


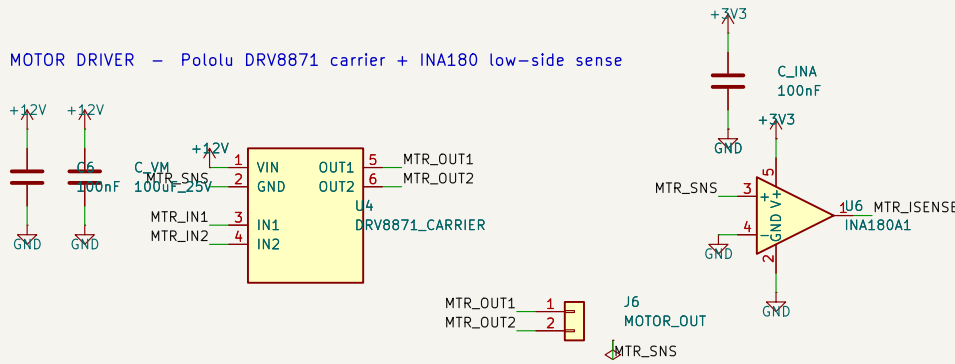
POWER NET FLAGS



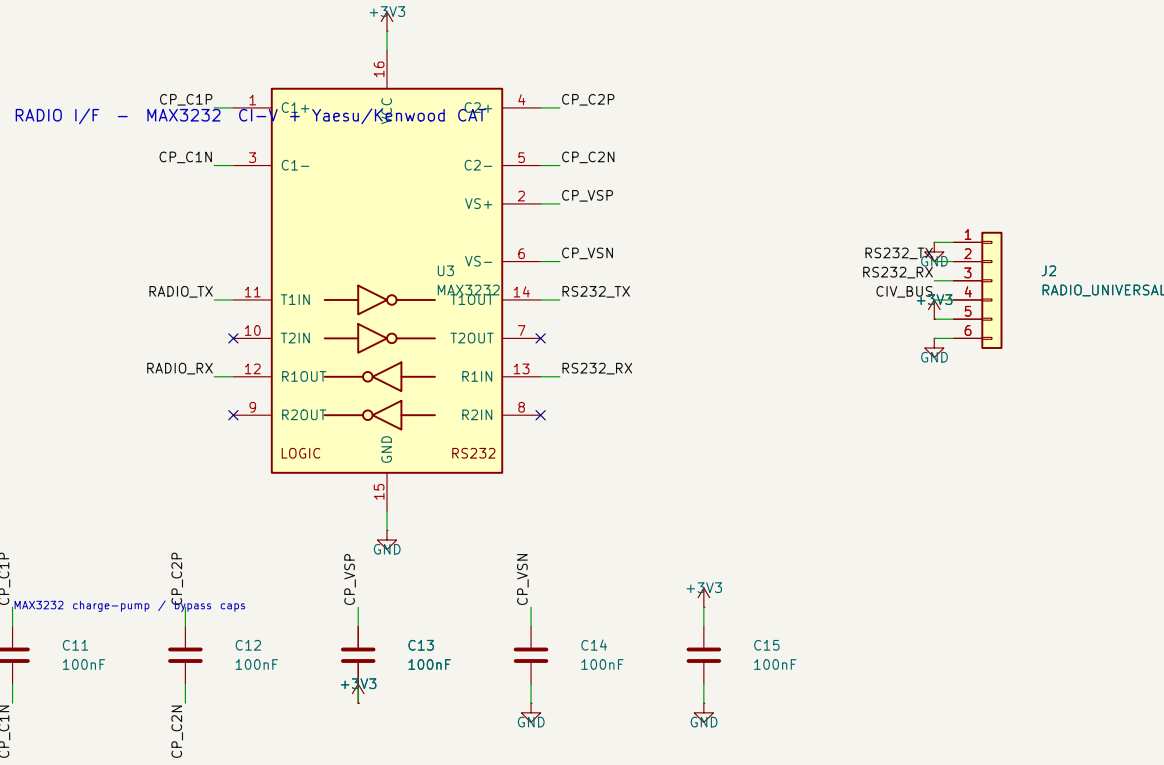
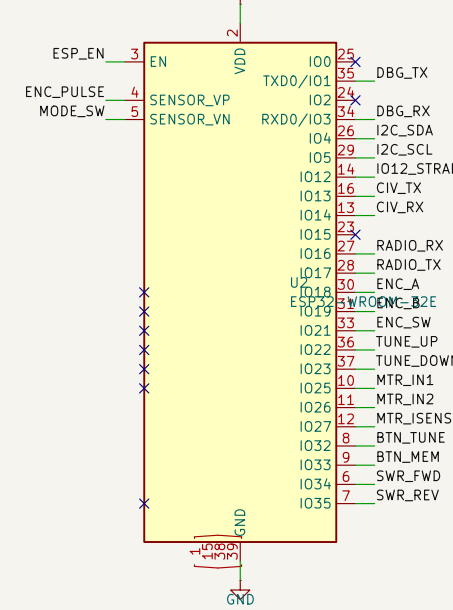
POWER SUPPLY - LMR33630ADDA 12V -> 3.3V 3A



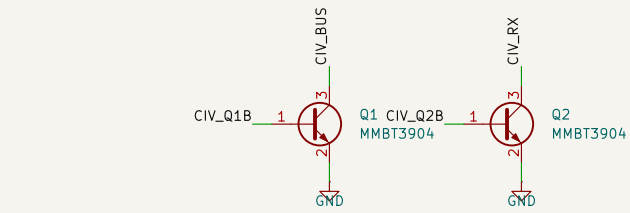
MOTOR DRIVER - Pololu DRV8871 carrier + INA180 low-side sense



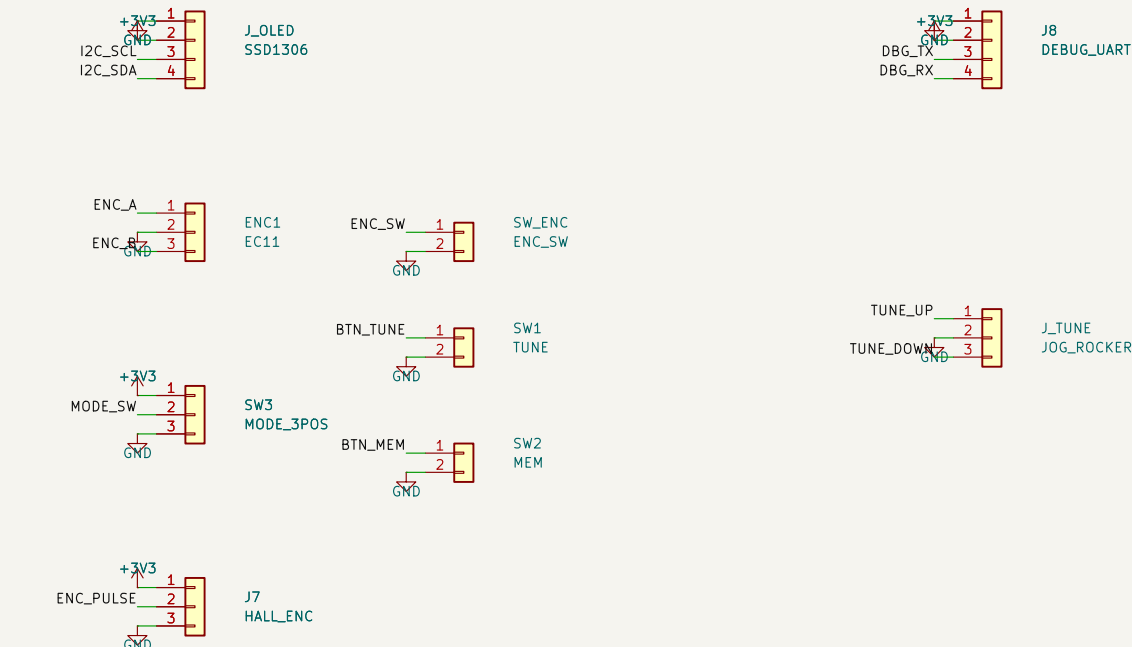
ESP32-WROOM-32E - main controller



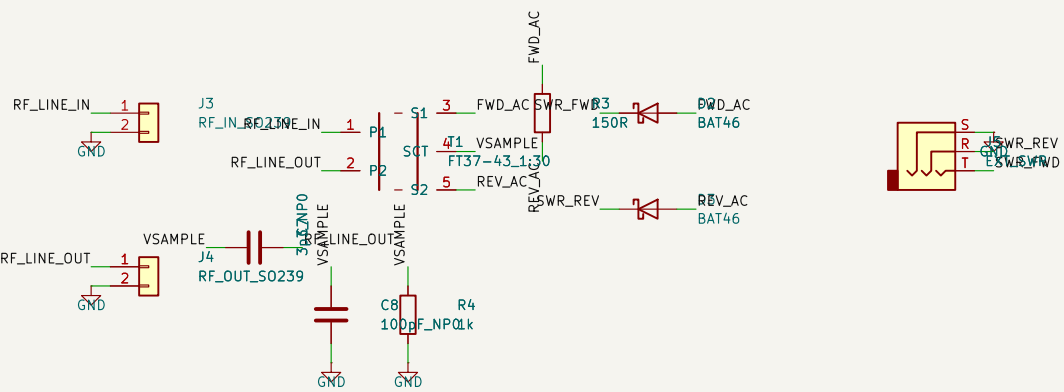
CI-V BUFFER (com) - shares DE-9 pink; invert UART1 in firmware



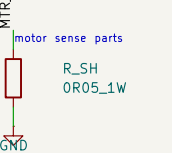
USER INTERFACE + I/O



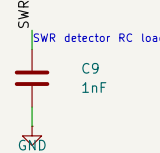
SWR COUPLER - Bruene bridge (CT FT37-43 + capacitive V-sample)



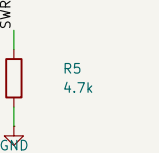
MTR_SNS



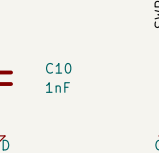
SWR_FWD



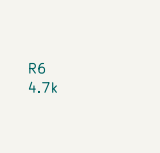
SWR_REV



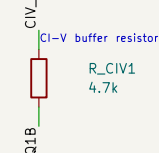
SWR_FWD



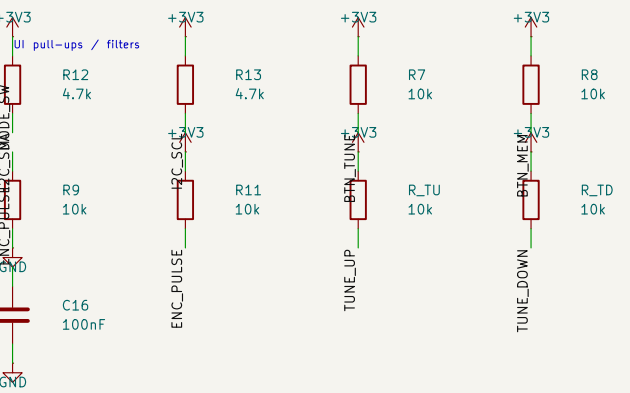
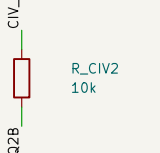
SWR_REV



CI-V_TX



CI-V_RX



Auto (CI-V/CAT) + semi-auto (SWR) + manual: label-driven netlist

Sheet: /
File: tuner.kicad_sch

Title: ESP32 Screwdriver Antenna Tuner

Size: A1 Date: 2026-06-14 Rev: 0.2

KiCad E.D.A. 10.0.3 Id: 1/1